



NTU Academy for Professional
and Continuing Education

(SCTP) Advanced
Professional Certificate

Data Science and AI



Attendance

Coding Class Norms

For a Smooth & Productive Session

- **Maintaining Class Flow**
 - Complete all pre-work — class builds on it.
 - If pre-work isn't done: listen, observe, catch up later.
 - If majority face the same issue, we will pause to address together.
- **Self Sufficiency**
 - Review instructions/materials.
 - Search online — Official docs, Google Search, Stack Overflow, AI assistants.
 - Attempt troubleshooting
- **Peer Learning**
 - Build relationship, collaborate with your buddy or assigned group to solve problems.
 - Supporting peers strengthens teamwork, deepens understanding,
 - Enhance learning from diverse skills and viewpoints

1.4 SQL Basics - DML

Module Overview

1.1 Introduction to Data Science

1.2 Introduction to Database

1.3 SQL Basic - DDL

1.4 SQL Basic - DML

1.5 SQL Advanced

1.6 Introduction to Numpy

1.7 Introduction to Pandas

1.8 Exploratory Data Analysis (EDA) Basic

1.9 EDA Advanced

1.10 Data Visualisation

Recap of Previous Lesson

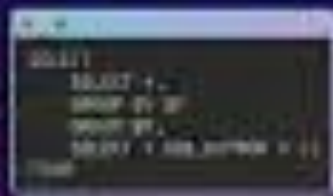
- Describe the following SQL DDL statements
 - CREATE
 - ALTER
 - DROP
 - UPDATE
 - COPY
- What are indexes useful for?
- What is the difference between a Table and a View?

Lesson Objectives

- SQL Data Manipulation Language (DML) statements
- Operators and Functions
- Filters and Sorting
- Aggregate functions and **GROUP BY**
- Advanced operators and functions

SQL Basics: Data Manipulation Language (DML)

Module 1.4 | Flipped Classroom Edition



Data Science



Learning Outcomes

By the end of this lesson, you will be able to:

1. **Retrieve** specific columns and rows from a database using `SELECT` , `WHERE` , and `ORDER BY` .
2. **Summarise** datasets using aggregate functions (`COUNT` , `SUM` , `AVG` , `MIN` , `MAX`) with `GROUP BY` .
3. **Filter** grouped results using `HAVING` and explain when to use `WHERE` vs. `HAVING` .
4. **Transform** data using `CASE` statements for categorisation and `CAST` for type conversion.